Speech **E**nabled **A**utonomous **L**iquid-handler

ME 326 SEAL Team 6

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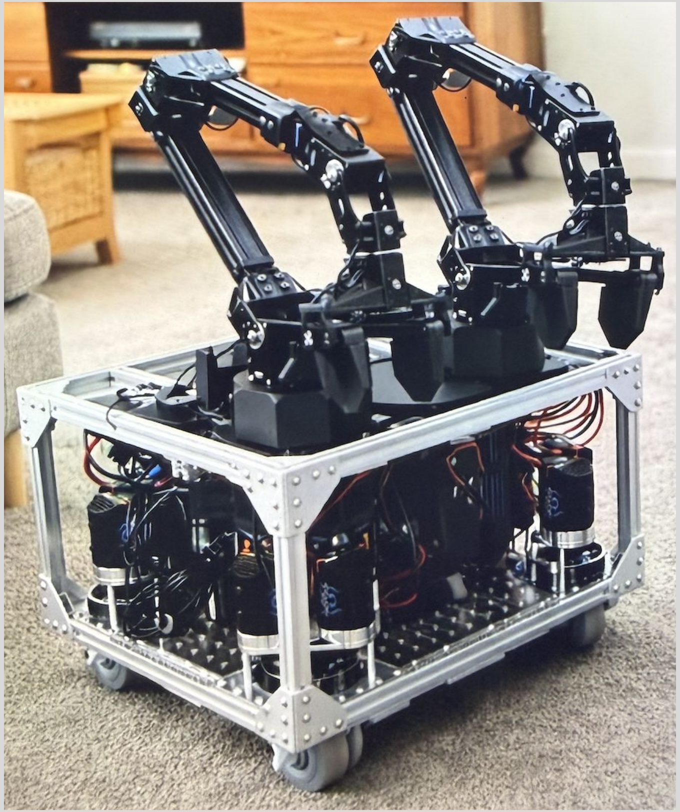
Meet the Robot



Meet the Team

Andrew Pasco

Yash Rampuria

Aditya Kothari

Luke Yuen

Ottavia Personeni

Mete Gumusuyak

Rohit Arumugam

Task 1

Approach: Rotate to find the banana, center it in the FOV, stop at distance *x*.

Compute: Generate a top-grasp pose using the banana's bbox/PC.

★ **Grasp:** Grab the banana, close the gripper, and lift to the sleep pose.

Return: Navigate to home.

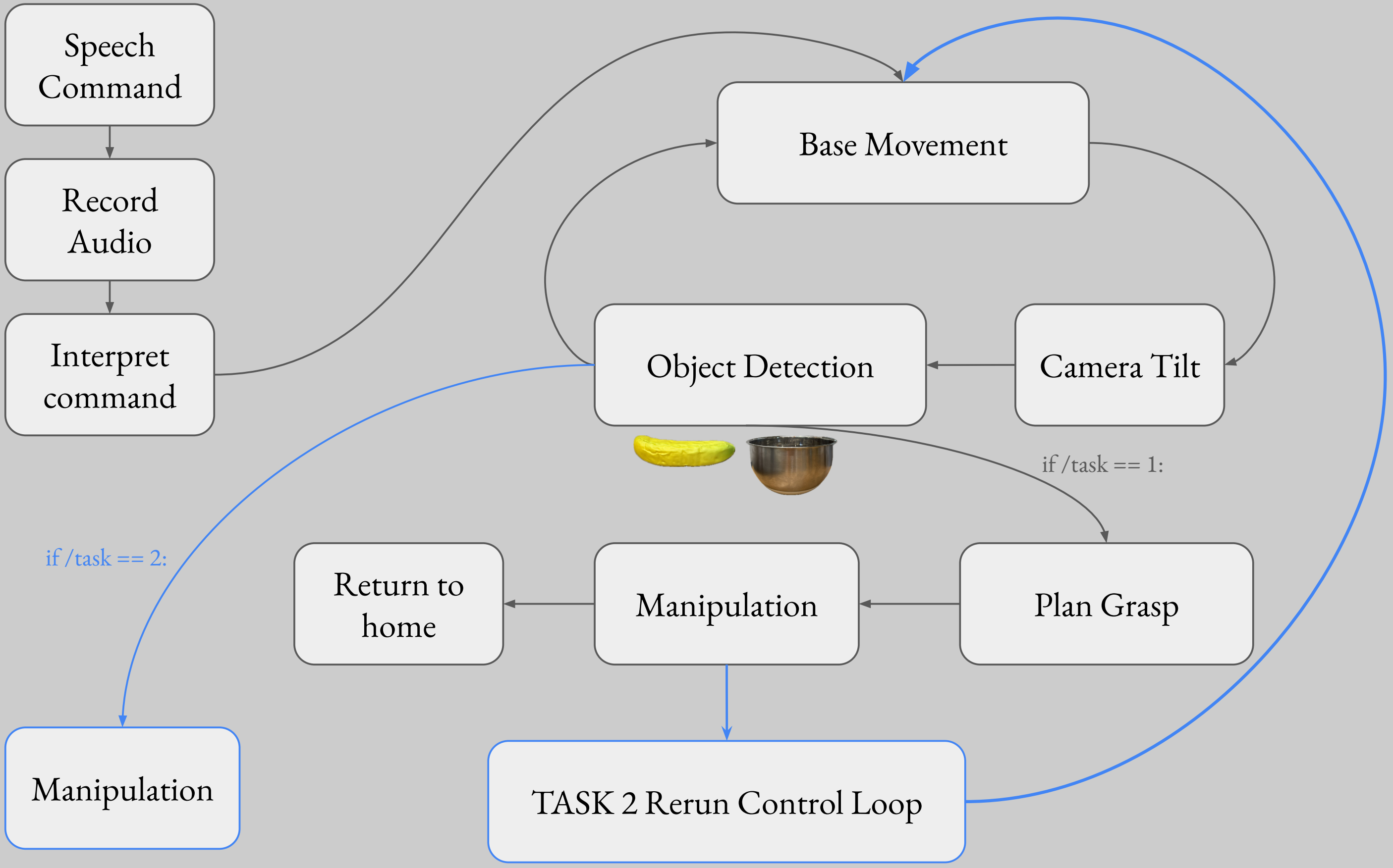
Task 2

…start from ★

Approach: Rotate to find the bowl, center it in the FOV, and stop at distance x.

Extend: Move the arm from the sleep pose to the forward pose.

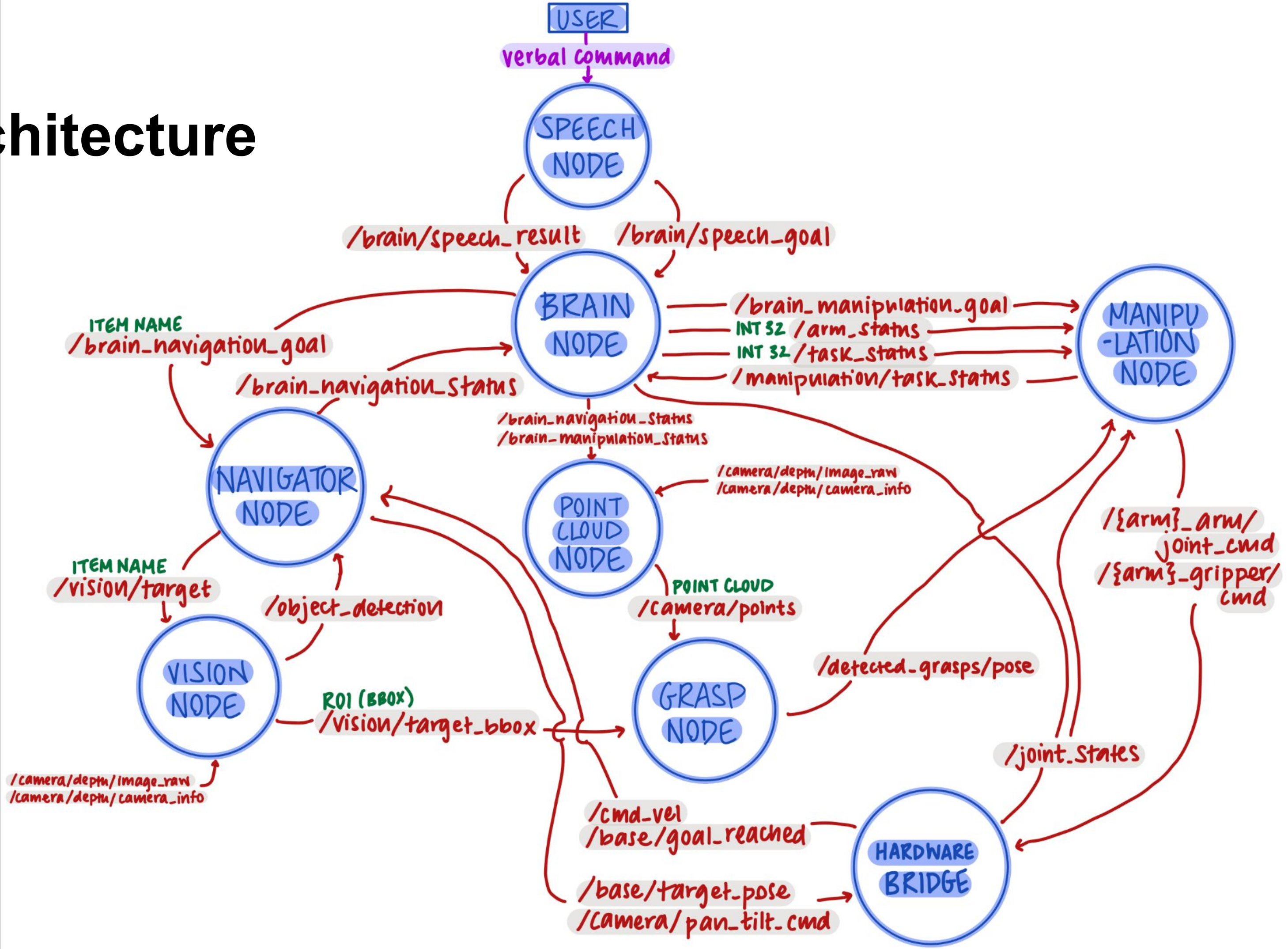
Release: Open the gripper to drop the banana



Software Architecture

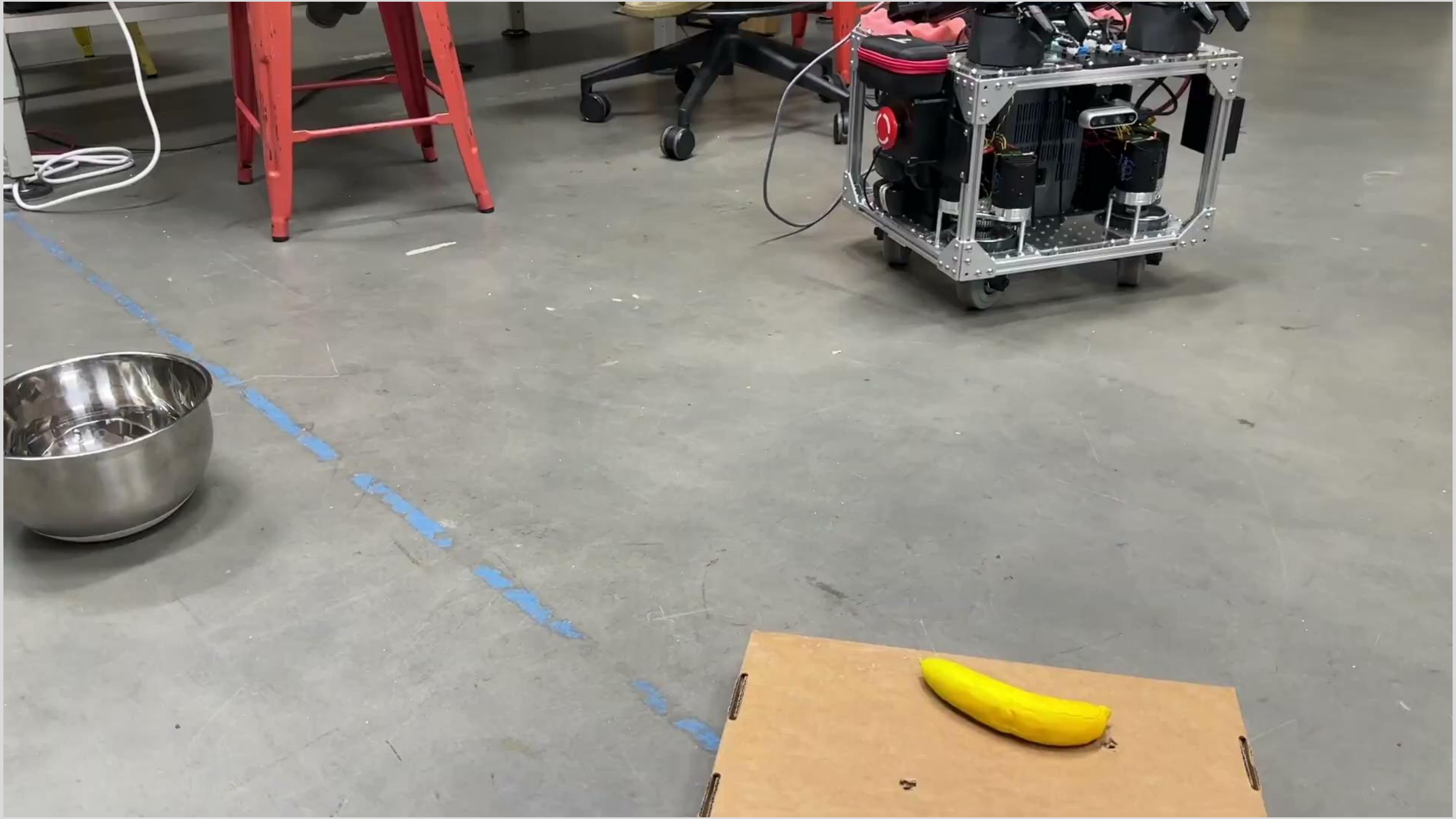
To see our team's full source code, please see the GitHub repo linked below.

[GitHub Repository]

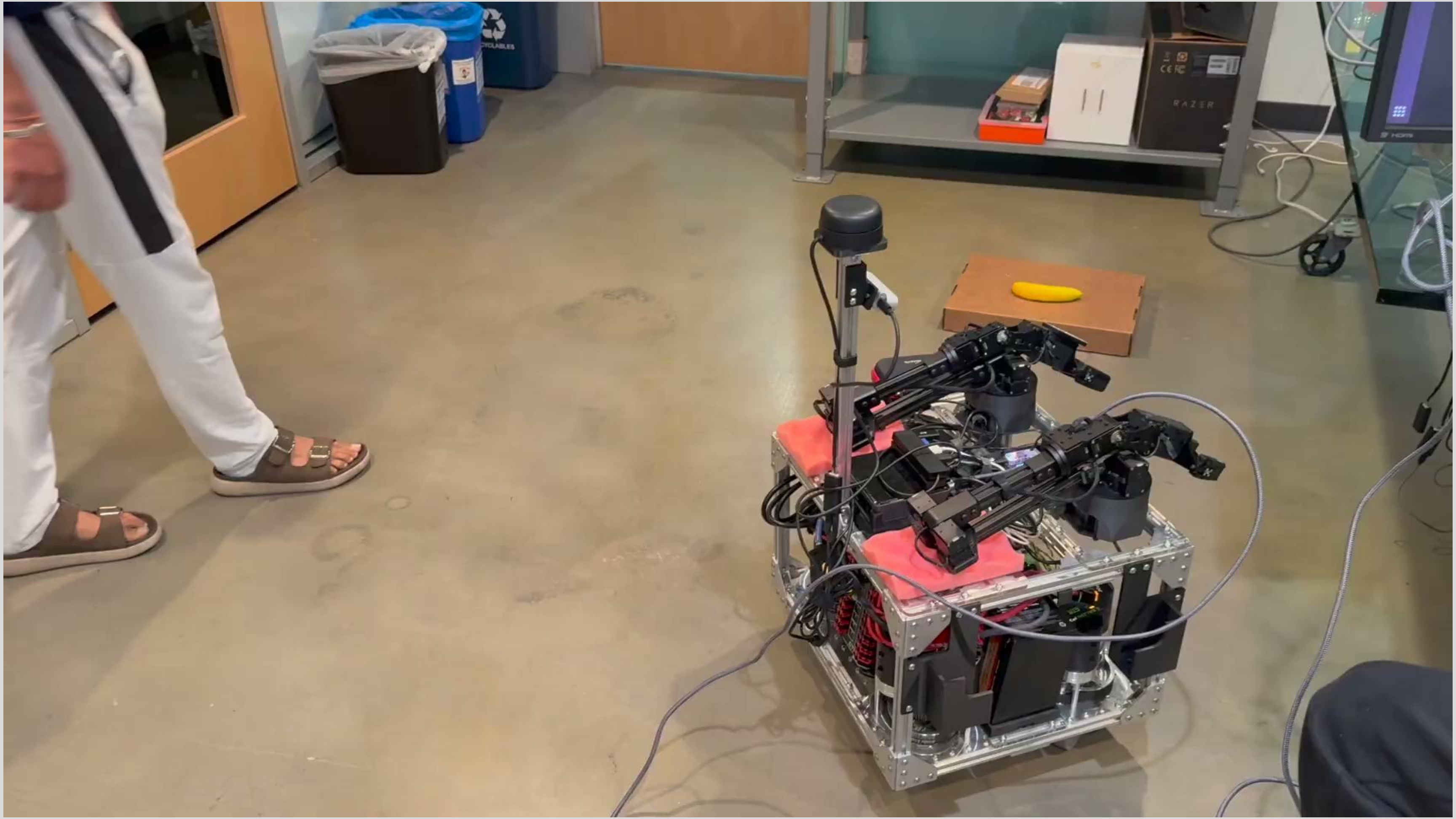


Task 1 Video

Task 2 Video



Task 2.5 Video



Task 3 Approach

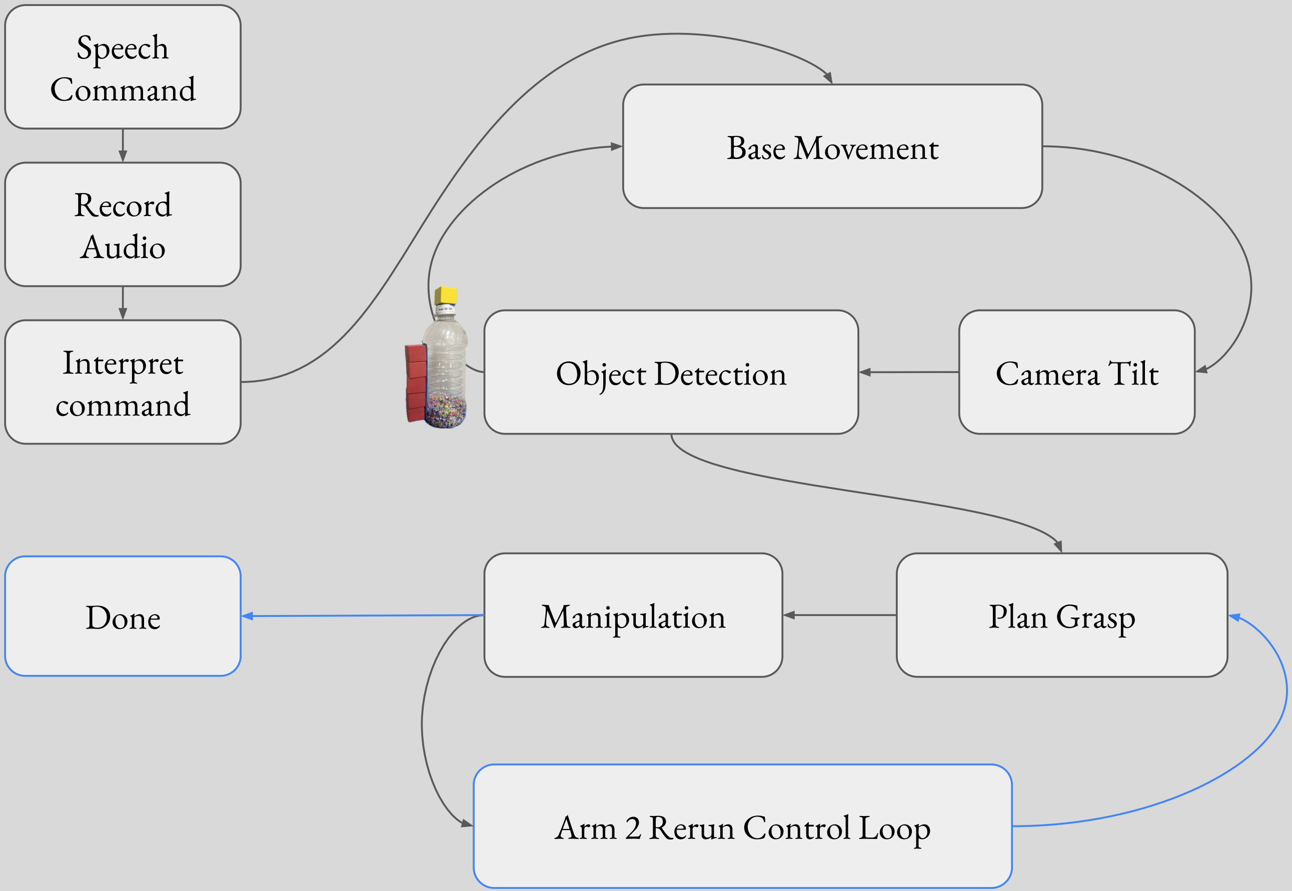
Search & Approach: Rotate to detect the bottle, center it in the FOV, stop at distance *x*.

Grip Bottle Body (Left Arm): Compute a side-grasp pose from the bottle's bbox/PC; close gripper.

Grip Bottle Cap (Right Arm): Compute a top-grasp pose from the cap's bbox/PC; close gripper.

Uncap: Twist right wrist 180° and retract the arm to its sleep pose.

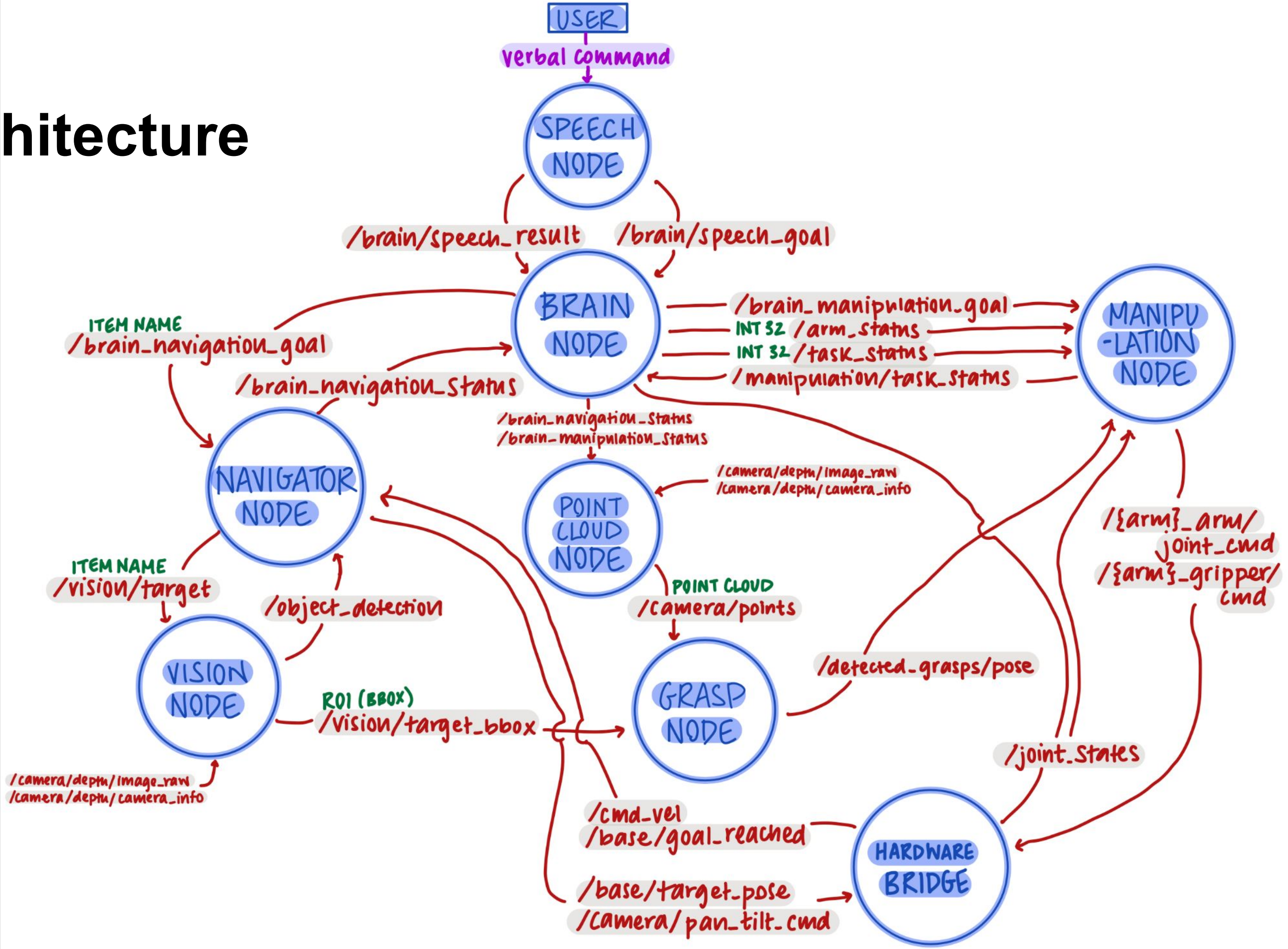
Pour: Raise and rotate the left arm to empty the beads.



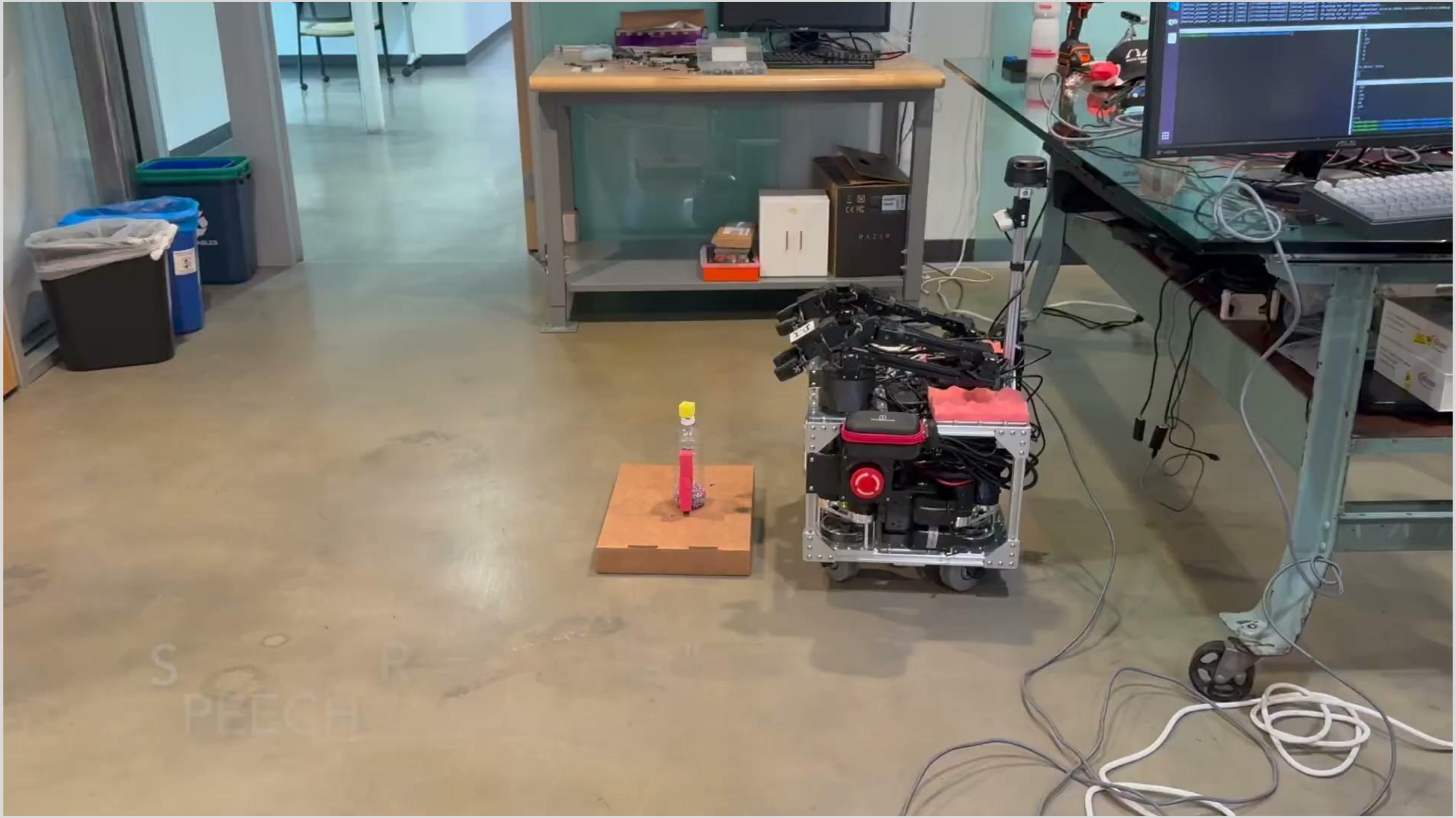
Software Architecture

(same as task 1/2, designed to be **modular**)

The Brain Node publishes /arm_status, /task_status flags to configure downstream nodes, allowing the same code to **serve all three tasks without modification**.



Task 2.99 Video



Conclusion

Issues & Room for Improvement:

- Robust grasp pose generation for varied objects
- Improved/altered architecture to resolve camera firmware inconsistencies
- Lower-latency segmentation (lighter-weight local model)
- More realistic implementation of task 3 (wider grippers to accommodate bottle without need for a blocky “grip”)

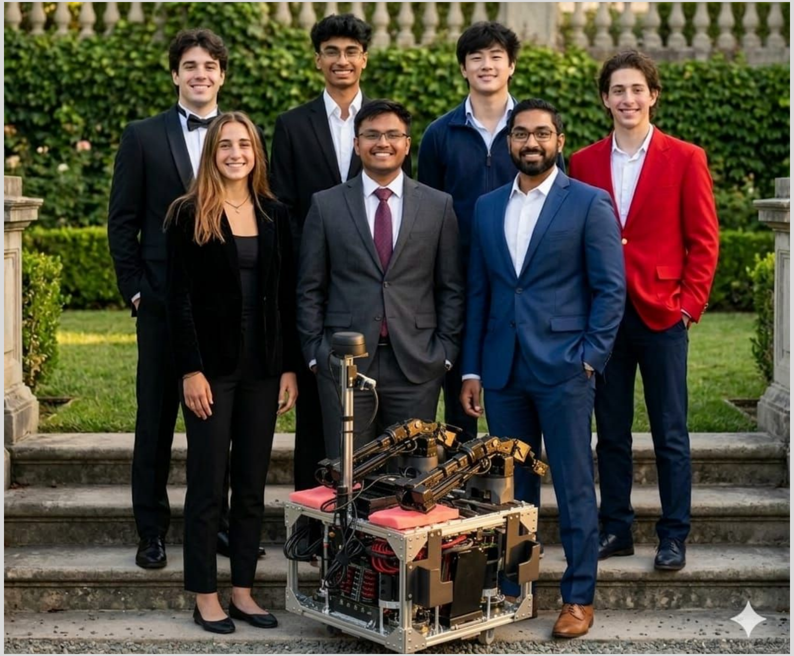
Future Work & Applications

- Geometric frontier-based exploration rather than reactionary navigation
- Act as a Barista/Bartender
- Assist Elderly and Specially-Abled persons with object retrieval tasks
- Precise pouring of medicinal doses for patients

Thank You!

To all the TAs for your hard work throughout the quarter and helping us stay sane!

And to Prof. Kennedy for a great quarter of collaborative robotics!



(and to the team whose chocolates we stole 🙃)

Approach

TASK 1

- Robot rotates in place to detect the banana in its FOV
- It then navigates to banana, while keeping it centered in FOV
- At *x* distance from the banana, robot stops
- Grasp node uses banana bbox/PC to output a grasp pose
- Manipulation: use grasp pose to grab banana from the top

★ Close gripper around banana and lift it back to sleep pose

- Navigate back to home

TASK 2

- From ★, robot rotates in place to detect the bowl in its FOV
- It then navigates to bowl, while keeping it centered in FOV
- At *x* distance from the bowl, robot stops
- From sleep pose, extend arm out to forward pose
- Open gripper to drop banana

Task 3: Approach

- Robot rotates in place to detect the bottle in its FOV
- It then navigates to bottle, while keeping it centered in FOV
- At *x* distance from the bottle, robot stops
- Grasp node uses bottle bbox/PC to output a grasp pose for red block at side of bottle
- Use grasp pose to grab bottle from the side with left arm (close gripper)
- Grasp node uses bottle cap bbox/PC to output a grasp pose for yello block on top of bottle cap
- Use grasp pose to grab bottle from the top with right arm (close gripper)
- Twist right wrist 180º and lift right arm to sleep pose to remove cap
- Raise and rotate bottle with left arm to pour out beads